

Lesson 5: Libraries Explore

45 minutes

Overview

Students learn the basics of libraries while building upon the envelope model of a function with a folder to represent a library.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-3** - Programmers break down problems into smaller and more manageable pieces. By creating procedures and leveraging parameters, programmers generalize processes that can be reused. Procedures allow programmers to draw upon existing code that has already been tested, allowing programmers to write programs more quickly and with more confidence.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Preview Libraries

Activity (35 minutes)

Wrap Up (5 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Clearly write documentation for functions in a library
- Explain the process of preparing a function to be added to a library
- Use appropriate vocabulary to describe libraries

Preparation

- Review the slides and click through all the animations
- Watch the Libraries video which introduces the Libraries tool in App Lab
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 7 - Lesson 5 - Libraries Explore** - Slides [▼ Make a Copy](#)

For the students

- **Libraries in App Lab** - Video (**Download**)

Vocabulary

- **API** - Application Program Interface - specifications for how functions in a library behave and can be used
- **Library** - a group of functions (procedures) that may be used in creating new programs

Teaching Guide

Warm Up (5 minutes)

Preview Libraries

 **Discuss:** *How could you share a function with another person so they could use it in their own program?*

Discussion Goal: Guide student answers towards possible answers like these -

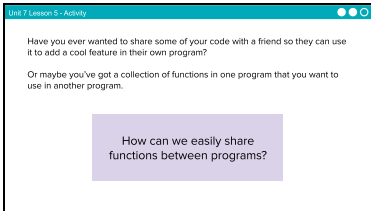
- email code to each other
- used a shared document
- copy/paste functions from one tab to another

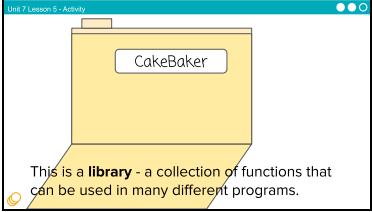
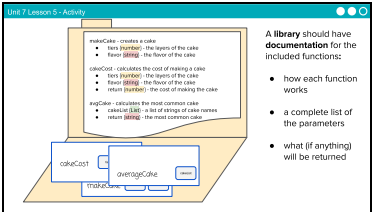
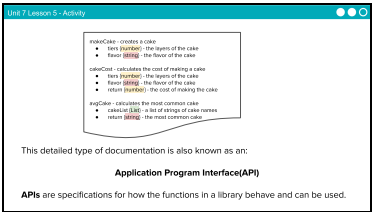
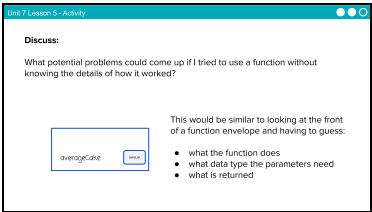
Remarks

Today we are going to learn a new way to share code.

Activity (35 minutes)

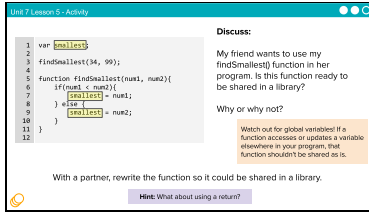
 **Display:** Use the activity slides for this lesson to guide the unplugged activity on Libraries.

Slides	Speaker Notes
	Say: Have you ever wanted to share some of your code with a friend so they can use it to add a cool feature in their program? Or maybe you've got a collection of functions in one program that you want to use in another program. Today we are going to explore the answer to this question: How can we easily share functions between programs?

Slides	Speaker Notes
 Unit 7 Lesson 5 - Activity CakeBaker This is a library - a collection of functions that can be used in many different programs.	 Click for animation: Stop once all three envelopes are on the screen. Say: These are functions. We recently learned about functions with parameters and return values.  Click for animation: Click until the library definition appears on the screen. Say: The envelopes have been grouped together in a folder. This represents a library. A library is a collection of functions that can be used in many different programs.
 Unit 7 Lesson 5 - Activity A library should have documentation for the included functions: • how each function works • a complete list of the parameters • what (if anything) will be returned	Say: Let's take a look at what's inside the library. If we open up the folder, we see the functions and some documentation. A library should have documentation for each of the included functions. The documentation should include: • how each function works • a complete list of the parameters • what (if anything) will be returned
 Unit 7 Lesson 5 - Activity This detailed type of documentation is also known as an: Application Program Interface(API) APIs are specifications for how the functions in a library behave and can be used.	Say: This detailed type of documentation is also known as Application Program Interface(API). APIs are specifications for how the functions in a library behave and can be used. Discuss: With a partner, review the parts of the documentation.
 Unit 7 Lesson 5 - Activity Discuss: What potential problems could come up if I tried to use a function without knowing the details of how it worked? This would be similar to looking at the front of a function envelope and having to guess: • what the function does • what data type the parameters need • what is returned	Discuss: What potential problems could come up if I tried to use a function without knowing what it does or how to interact with it? This would be similar to looking at the front of a function envelope and having to guess: • what the function does • what data type the parameters need • what is returned Note: If students are struggling to come up with ideas, give a few hints: • what if there's an error when you run the code which calls this function? • what happens to a return if you don't store it in a variable or print it to the console?

Slides

Speaker Notes



Discuss: My friend wants to use my `findSmallest()` function in her program. Is this function ready to be shared in a library?

Click for animation

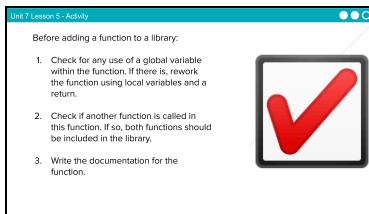
Say: Watch out for global variables! If a function accesses or updates a variable elsewhere in your program, that function shouldn't be shared as is.

Click for animation

Do This: With a partner, rewrite the function so it could be shared in a library.

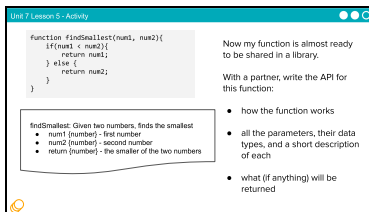
Click for animation

Note: The problem here, is that the function uses a global variable. Students should try to rewrite the function so only local variables and a return are used. The answer is shown later on in the slides.



Say: These are the things you should consider before adding a function to a library:

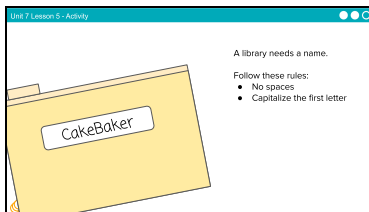
1. Check for any use of a global variable within the function. If there is, rework the function using local variables and a return.
2. Check if another function is called in this function. If so, both functions should be included in the library.
3. Write the documentation for the function.



Say: Here's the same function, now written to be shared in a library. How does it look compared to yours? It may look a little different - and that's ok, as long as the function works as expected. Now my function is almost ready to be shared in a library. With a partner, write the API for this function:

- how the function works
- all the parameters, their data types, and a short description of each
- what (if anything) will be returned

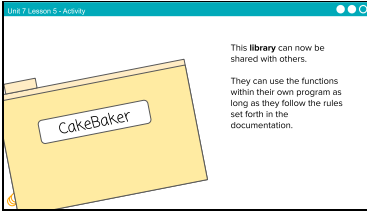
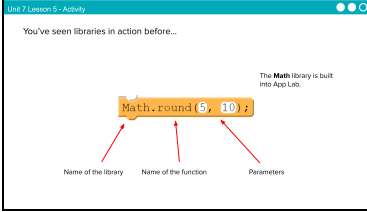
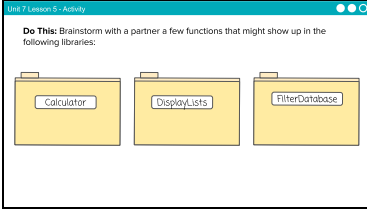
Click for animation When the class is ready, click through to see a possible answer.



Say: A library needs a name. For our libraries in this class, we are going to follow the following rules:

- No spaces
- Capitalize the first letter

Click for animation

Slides	Speaker Notes
	Say: This library can now be shared with others. They can use the functions within their own programs as long as they follow the rules set forth in the documentation.  Click for animation
	Say: You've seen libraries in action before. The Math library is built into App Lab. Notice the name of the library first, then after a dot, the name of the function and finally the parameters. This is the same formatting you will use to call functions in libraries: Library name, followed by function name, and finally the parameter(s)
	Do This: Brainstorm with a partner a few functions that might show up in the following libraries: Calculator, DisplayLists, FiterDatabase. Note: Here are some ideas: • Calculator: functions similar to what you would find in a common calculator • DisplayLists: functions that help you display a list on a screen in different ways, such as one list item per line or divided with bullet points • FilterDatabase: functions that allow you to filter a database for different things, like all the items that start with the letter "a"

Remarks

Now let's take a look at how libraries work in App Lab.

 **Video:** Watch the [Libraries in App Lab](#) video.

Wrap Up (5 minutes)

 **Journal:** Have students add the following words to their journals: library, API

Remarks

Great work today! Libraries are different than other programming concepts we've explored like variables and conditions. They offer a way to organize and share your code with others.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Evaluate if the following function is a good candidate to be placed in a library. Why or why not?



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